NATHAZ AI LAB / TECHNICAL REPORT · EXPLORATORY TRIAL

R23.4c Thunderstorms: AI forecasts driving WRF

Group AI Weather & Climate Models for Wind Engineering

Status Shelved

Period 2026-10-14 14:38 — 2026-10-17 10:43

Stage conclusion

Project report

Thunderstorm: AI Forecast Driving WRF (K)

Objective

Using the exact same convective-permitting WRF setup (4 km single domain, no cumulus parameterization), drive the initial and lateral boundary conditions with Pangu forecasts vs operational GFS forecasts at 24/48/72 h lead times across observed historical thunderstorm events already simulated by the WRF line, comparing storm initiation timing/location, radar reflectivity objects, and station wind gusts to judge whether Pangu-driven WRF is non-inferior to GFS-driven WRF. Serves as the physical downscaling accuracy benchmark for Node L.

Current Status

10-17 Shelved: None of the three lead times (24/48/72 h) met "13-level Pangu driving is non-inferior to full-level GFS driving" (0/3), and both groups had 24 h position errors >150 km, making "worth continuing" unreachable under the frozen rules. Single-event, qualitative; model and vertical level count are not decoupled, so we do not claim "AI driving is inferior to GFS." Restarting requires running R24 (13-level GFS) first. No jobs currently running.

Decisions Made

- 2026-10-17 Shelved ("worth continuing" not met): 0/3 non-inferior across 24/48/72 h and both 24 h position errors >150 km; do not state "AI driving is inferior to GFS" (model and level count are not decoupled, and R24 was not run); see decision for restart conditions →
- 2026-10-15 Protocol refinement (filling an unstated edge case while keeping criteria unchanged): set the formal minimum area threshold for `≥40 dBZ` radar objects to `≥64 km²` (reporting unthresholded and `≥16 km²` alongside), and shift the onset-time search window start back to `00Z` (while keeping the `03–09Z` spatial evaluation window); refined per review verdict in `v2`: count a vote only when the `≥16 km²` and `≥64 km²` area thresholds agree (counting neither a positive nor a negative vote if they disagree), apply the area threshold uniformly across all object metrics, measure `"within R km"` to the nearest object grid cell, and evaluate left-censored onset times (when an object is already present in the `00Z` frame) via interval bounds without awarding a positive vote if ambiguous →
- 2026-10-14 Design `v2`: switch the GFS baseline to full vertical levels (per Deng 2025 abstract showing that truncating GFS to 13 levels degrades skill most severely), moving 13-level GFS to optional sensitivity run `R24` →
- 2026-10-14 Experimental design `v1`: initialize primary Pangu runs from GFS analyses (matching the GFS group's initial state) while keeping ERA5-initialized Pangu as optional `E24`; use identical 6 h boundary intervals and land-surface fields, and add the 6 h pair (`G06/P06`) as a pipeline anchor and gate →
- 2026-10-14 Node established: first case is the Akron `2020-06-09` macroburst (with Wister `2018-08-10` as an optional initiation control), running WRF on CRC (0 SU charged) →

Attempts and Results

Date	What Was Done	One-Line Result
2026-10-17	WRF line delivered 72 h pair (formal G72/P72 dt15, CRC 1571882; dt20 auxiliary column 1560832) and completed 150 km existence; Lead tallied votes	All four 72 h criteria agreed: P72 position error 307.7/299.7 km vs G72 104 km, and at ≥64 km² G has an object while P has none → not non-inferior; all three lead times (24/48/72 h) scored "clearly worse" (48 h on timing alone); drafted "Shelved (AI driving inferior to GFS, single-event qualitative)" per decision rule and sent to 6-reviewer panel before changing status
2026-10-15	WRF line delivered formal v2 table for G48/P48 (job 1552228)	At ≥64 km²: G48 first appears at 00:30Z (−160 min), 06:10Z centroid error 316.8 km, object area 80 km², max 10 m wind 17.1 m/s; P48 has no onset within 50 km during the window (right-censored), centroid error 273.3 km, area 287 km², max wind 17.0 m/s; at ≥16 km², G48 is −140 min / 275.2 km and P48 has no onset / 174.7 km. CSV verified by Lead; requested 150 km existence column across pairs; final interpretation awaits the 72 h pair (~10-18)
2026-10-15	WRF line accepted 72 h time-step pairing	Formal 72 h pair = G72 (dt=15 s) vs P72 (dt=15 s) (job 1547482 running); G72 (dt=15 s) staged in a clean CRC directory and submitted after WRF-line review, expected to finish late on 10-17; completed G72 (dt=20 s) (1544570) retained as auxiliary time-step sensitivity column; 48 h pair delivered on 10-15 evening
2026-10-15	WRF line delivered formal v2 table for G24/P24 (job 1548771)	At ≥64 km²: observed onset 03:10Z, G24 04:00Z (+50 min), P24 00:20Z (−170 min); 06:10Z centroid error G24 183.4 km / P24 307.9 km; max 10 m wind 22.2 / 21.5 m/s. At ≥16 km²: P24 is left-censored (present at 00Z frame), selecting a separate 32 km² object at 233.1 km. CSV verified by Lead; because P72 hit a CFL blow-up at dt=20 s and switched to dt=15 s, requested G72 rerun at dt=15 s

Date	What Was Done	One-Line Result
2026-10-15	WRF line delivered formal v2 table for G06/P06 (job 1545333)	At ≥64 km²: observed onset 03:10Z, G06 04:40Z (+90 min), P06 00:50Z (−140 min, inside spin-up window, anchor only); 06:10Z centroid error G06 53.1 km / P06 66.6 km; object area observed 1,380, G06 928, P06 1,055 km²; max 10 m wind within 100 km is 24.8 / 26.7 m/s vs observed gust 37.0 m/s. G06 passed the pipeline gate under v2 (≥64 km² object within 150 km). At ≥16 km², P06 selects another 48 km² object. IEM N0Q cross-check confirmed spatial overlap with different area scaling; accepted by Lead
2026-10-15	WRF line adopted Protocol v2	Updated evaluation script (preserving v1 script) to apply area filtering prior to scoring hit50 / 150 km / gate; re-verified G06 gate at ≥64 km²
2026-10-15	WRF line adopted two protocol refinements	Configured formal table across 3 area thresholds × 2 window start times; noted that at 06:10Z the initial 3-radar composite had only a single 2 km pixel ≥40 dBZ and cross-checked against IEM N0Q
2026-10-15	Research assistant generated driving fields v2.2 (CRC job 1544281, 0 SU)	All 66 files passed SHA256 verification (spot-checked by Lead); only Q2 changed while all other fields remain byte-for-byte identical to v2.1; within 300 km of KAKO, v2.2 − v2.1 is −0.37 to −0.91 g/kg, matching the exact opposite sign of the moist bias reported by the WRF line
2026-10-14	WRF line G06/P06 preprocessing trial (jobs 1542899/1542900)	metgrid passed (G 32 levels / P 14 levels); confirmed open-water SNOW=0, SMOIS=1, SST=TSK; real.exe produced lateral-boundary vertical interpolation NaNs at 06-09 12Z in both groups (identified by WRF line as an internal configuration issue bypassed via all_ic_times=true); Q2 sensitivity check showed v2.1 was +0.38 to +0.91 g/kg too moist (max point diff 1.6–9.3 g/kg, exceeding the 1 g/kg tolerance) → triggered v2.2

Date	What Was Done	One-Line Result
2026-10-14	Research assistant generated 8 driving-field sets (`v2.1`, `0.25°` GFS)	66 WPS intermediate files (1.6 GB) passed readback and SHA256 verification (`G` 31 pressure levels / `P` 13 levels, land-surface fields byte-identical between groups; ocean soil/snow nearest-neighbor filled and Pangu negative `q` clipped to `0`, all labeled as assumptions)
2026-10-14	Driving-field archive pre-check by research assistant	Historical 2020 GFS `0.5°` multi-lead forecasts are no longer publicly archived (NCEI returns 404 beyond ~2 years; AWS retains only 30 days), whereas native `0.25°` GFS forecasts are downloadable from NCAR GDEX `d084001` (0 CU)
2026-10-14	Grok quick search for precedents	Found zero prior studies driving convection-permitting models with AI weather forecasts for mid-latitude downbursts / severe thunderstorm winds; verified Deng 2025 abstract (33-level GFS ≈ 13-level Pangu, whereas 13-level truncated GFS is worst) and Brewer 2025 abstract (GraphCast-WRF comparable to GFS/ECMWF-WRF for downslope windstorms)

Parameter Sources

Parameter	Value	Source or "Assumption"	Places Checked
WRF `d01` configuration	`4 km`, `748×600×80`, `dt = 20 s`, `cu=0`, `mp=8`, `PBL=2/sf=2`, `lw=4`	Original author `namelist`	WRF Lead reply
Compute cost	~1 SU / model hour (unbenchmarked)	Scaled by WRF Lead from `exp-A`	Pending short test
Map projection	Lambert conformal, `38N/100W`, `truelat = 38/38`	WRF line; original `wrfinput` metadata	—
`grid_fdda`	`0`	Original `namelist.input` line 113	—
GFS archive for this experiment	Native `0.25°`, NCAR GDEX `d084001`	Pre-check by research assistant (`README`)	NCEI, AWS, and `d335000` lack 2020 `0.5°` forecasts

Parameter	Value	Source or "Assumption"	Places Checked
RH → q (Group G), Pangu q < 0 clipping, ocean soil/snow nearest-neighbor fill	Documented in preprocessing spec	**Assumed** (implementation choices)	Verified via metgrid/real
2 m specific humidity (Q2)	v2.2: take q from the lowest complete pressure level with GHT > SOILHGT (v2.1 ln p extrapolation voided)	**Assumed** (Design v2.2 based on WRF-line Q2 sensitivity check; no literature basis)	Checked Brewer repo (rfovell/GraphCast-WRF, which omits 2 m RH)
Original paper's driving data	GFS 0.5°, 00Z cycle, 3-hourly	WRF line	—
Observational verification source	NEXRAD Level-II column-maximum reflectivity composite + KAKO METAR/SPECI, anchored to 37.04 m/s peak gust	WRF line on 10-15 (1-minute ASOS missing on that date; NCEP archive lacks MRMS on that date)	Evaluation thresholds unchanged
Observed peak timestamp	~06:10 UTC (02:10 EDT at KAKO)	Card GZYPXN4S PDF pp. 9–13 (cited via WRF-line REPORT_FINAL)	Pending direct verification by this node
GFS baseline vertical levels	Full 32 pressure levels (not truncated to 13)	Deng 2025 *AOSL* 10.1016/j.aosl.2025.100589 abstract (verified via OpenAlex)	Full text behind ScienceDirect 403
Pangu initial condition	Same-cycle GFS f000 analysis	Unstated in Deng 2025 abstract (Grok claimed SI specifies GFS analysis — **unverified**)	Pending full-text check
Surface/soil fill, 2 m humidity, PSFC	Land surface copied from initialization-time GFS f000; 2 m humidity derived identically across both groups; PSFC omitted	**Assumed** + WRF Lead guidance; cf. Brewer 2025 repo rfovell/GraphCast-WRF (Apache-2.0, latest push 2024-12-19)	Delegated check
WRF version	4.0.3	WRF Lead (results are sensitive to WRF version)	—

Open Items and Next Steps

- ideas delivers driving fields → verify field list and parity → formally notify of the delivery location.
- Grok quick check returned (see Attempts and Results). Deep Research step 2 **not done**: 30-day quota 30/30 exhausted; will issue once quota resets or Liang Hu adds quota.
- WRF Lead accepted v2: conclusions must explicitly state that the main comparison difference includes both model and vertical level count; if capacity permits, run R24 first and then E24; run one metgrid/real trial each for G06 and P06.
- Driving fields v2.2 delivered (10-15 05:00 --report to parent node, forwarded via root Lead to WRF Lead).

- WRF track started runs on 10-15 04:58 after re-verifying v2.2: boundary NaNs in real bypassed using all_ic_times=true (tested to have bitwise zero impact on lateral boundary forcing); running G06/P06 first (~half a day) and evaluating the G06 gate per Section 5; once past the gate, the three 24/48/72 pairs (6 integrations) will run in parallel (the longest 78 h group takes ~16–18 h wall-clock time, excluding queue wait). 3D fields retained only for 03–09Z, targeting <15 GB. Numerical tables saved and returned to this node; this node will interpret results per the frozen criteria and report to the parent node. Fallback follow-up at 10-16 16:41.
- Waiting on: design objections, delivery location, schedule.
- WRF track short test → 06 pair passes gate → 24/48/72 pairs → comparison table → conclusion (before presenting to Liang Hu

Boundaries

- Testbed rules: no single-experiment limit on Colab (corrected by parent project on 10-14 per Liang Hu's exact words "use Colab heavily at this stage, unlimited CU", superseding the previous ≤20 CU cap), but if a single experiment is expected to exceed 100 CU, pause and report to the parent node Lead first; must run colab stop when finished so sessions do not sit idle, and dispatch only one executor per round; supercomputer single job ≤50 SU (CRC does not bill SU), WRF jobs submitted by the WRF track bot, requiring ComputeOps resource request, registration, and separate commands for pre-check and submission; data from other tracks are read-only.
- Storage and compute locations (tightened by Liang Hu on 10-14): the workstation does not run computations or store compute data, holding only scripts, small results <200 MB per item, figures, and reports; if temporary storage >1 GB is strictly needed, report to the parent node Lead first. Driving fields and WRF outputs are placed only on Colab or clusters (currently on CRC); new large datasets prioritize TACC work or scratch storage (CRC group quota is at ~95%), and CRC defaults to a single node. Every task brief must state where computations run and where results are stored.
- Hierarchy (set by root Lead on 10-14, 5-level cap): this node is a level-2 node and **will not open level-3 sub-projects**; when subdivision is needed, use tasks within this node or report to the parent node Lead (Thunderstorms and Convective Scale) to open a peer level-2 node.
- Only two hops remain downward: this node → bot → tool. Do not allow tool bots (gpu-gemini, reviewer, litwiki, etc.) to delegate further in task briefs.
- The subtree list is not grouped by parent node; the UI project tree is authoritative.

Literature Basis

- Akron mechanism: card 2DPBW6VX PDF pp.6, 9–11; observations: Childs 2021 card GZYPXN4S Table 2, PDF p.16 (cited via WRF Lead, pending verification by this node)
- Xu 2024 ERL 10.1088/1748-9326/ad41f0; Liu 2024 JGR:MLC 10.1029/2024JH000207 (no card created, not in local library)

- Deng 2025 AOSL 10.1016/j.aosl.2025.100589 (abstract verified, no card created); Brewer 2025 WAF 10.1175/WAF-D-24-0097.1 (abstract verified, no card created) — these 4 papers have been delegated to litwiki to create cards and verify full texts (10-14 report back: PDFs not yet retrieved, forwarded to Literature to fetch full texts and create cards upon retrieval); do not cite in conclusions before cards are created.

Liang Hu Notes

- 10-14: Fundamentally different in scale from typhoons (B3); listed separately.

Key figures